SAMPLE

Great Observatories: NuSTAR

Nuclear Spectroscopy in Space

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Introduction

Of all the light the universe sends, the hardest kind spent the longest uncaught. The X-rays, the light that carries the fingerprints of the universe's most violent places, the accretion disks of black holes and the shock fronts of exploding stars, arrive at the Earth in two bands, and the two bands had very different histories: the soft X-rays, up to about ten kilo-electron-volts of energy, were focused and imaged by the great X-ray observatories of the late twentieth century, their mirrors building the pictures that made the field; but the hard X-rays, the band from a few to nearly a hundred kilo-electron-volts, resisted the mirror-maker's art, their radiation too fierce to reflect at the grazing angles the old coatings allowed, and the band was read for decades only through the collimators and the detectors, the light counted but never pictured.

NuSTAR, the Nuclear Spectroscopic Telescope Array, the subject of this book, is the instrument that ended the exclusion. Launched by NASA in 2012 as a small Explorer mission, NuSTAR was the first telescope in history to focus the hard X-ray band: its nested mirrors, coated with depth-graded multilayers that reflect even the fierce light at a grazing angle, its ten-meter mast, the longest deployable structure ever flown as part of a telescope, and its pixel detectors together made the first true images of the sky in the light the old missions could not focus, and the band opened. The mission's mass, its cost, and its size belong to the small-explorer class, the modest tier of the space program, and its achievements read like the work of a flagship: the mapped disks of black holes, the census of the hidden giants, the decay-mapping of fresh supernova debris, the radius-weighing of neutron stars, and the first pulsating ultraluminous source, a discovery that upended a standing interpretation of a whole observed class.

This book follows the mission whole. It begins with the band's history and the mirror problem, the decades of the uncatchable light; it follows the mission's concept and construction, the multilayer coating and the mast that were the enabling inventions; it surveys the observatory's workings, the orbit and the observing program, the calibration and the operations; it presents the science, the black hole mappings and the supernova chronologies, the compact-star measurements and the solar flares; and it closes with the mission's standing in the era it opened, the successors and the legacy, the small telescope that gave the universe's hardest light its own eyes.

A note on the book's sources and their limits, before the mission's story starts. The account rests on the mission's public record: the project's own technical publications, the optics papers and the calibration reports; the science papers of the discovery harvest, the spin measurements and the remnant maps, each cited as the literature states it; and the program's own documents, the selection records and the review histories, the mission's institutional paper trail. The numbers quoted, the band's energies and the mast's length, the launch's date and the program's milestones, are the record's own established facts, and where the field's interpretation stands open, the obscured-census debates and the ULX class's mixed nature, the account says so plainly rather than choosing for the reader. The mission's vocabulary, the multilayer and the grazing angle, the reflection spectrum and the decay line, is defined at each term's first use, and the reader who finishes the chapters holds the band's own language as well as its story, the small telescope's whole account in plain words.

The reader may also hold, before the technical account begins, the mission's place in the longer history of the telescope itself. The instrument that Galileo turned upward was a tube of glass; the great reflectors of the next centuries were mountains of polished metal; the space age added the orbital observatory, the telescope freed from the air; and each generation's instrument opened a band the previous one could not see. NuSTAR belongs to the latest of these openings, the last of the X-ray band's windows, and its story is therefore one chapter of a very old story, the history of the species' extending eyes. The chapters that follow tell the chapter's own details, the coatings and the mast and the harvest, but the frame is the older one: every band once had its "impossible," the light that no mirror could serve, and every band's impossible has fallen so far to the opticians' patient invention, the hard X-ray's turn arriving on a small rocket in the second decade of the new century, the pattern's most recent proof that the telescope's history is the history of the word "impossible" losing.

One further note on scope, the reader's map of what follows. The eleven sections of this account are built the way the series builds them: the mission's discovery and construction first, the characteristics and the working science second, the contexts and the standing results third, the honest difficulties fourth, the future and the lighter lore last, and the whole arranged so that each chapter stands alone for the reader who dips and reads in order of interest. The technical terms are defined at first use and reused deliberately; the mission's milestones are dated where the record dates them; and the account's own judgments, the meanings this introduction has promised, are gathered plainly at the end rather than scattered as assertions through the chapters' evidence. The reader who wants the mission in one sitting can take the chapters in order, and the reader who wants the answer to a single question,

what the mast was for, or what the pulsar discovery changed, can find it in its chapter's own paragraphs: the book is a tool as well as a story, and its organization serves both uses, the whole hard-band opening told once, completely, and in the order the understanding builds.

A last note before the start, the standing question the reader may bring: why hard X-rays at all, and why the band's opening mattered to anyone but the specialists. The answer the chapters give in full, but the short form belongs here: the band's light is the fingerprints' light, the signatures of matter at its most extreme, and the universe's deepest questions all pass through it. What is a black hole's spin, and how did it grow; what makes a star explode, and what does the ash's arrangement say; how dense is the matter in a neutron star, and what does that say about the atom's own heart; how does the Sun throw its flares, and the galaxy hide its growing giants: each answer, at some essential step, is written in light the soft mirrors cannot focus, and the decades of the collimated band read the words without the sentence. NuSTAR made the sentences readable, and the reader who completes this book will hold both what the sentences say and why the reading was hard, the whole account of the band's first focused eye and the violent universe it was built to read.

The reader should hold from the start the mission's central meaning: NuSTAR is the proof that the frontier of a field can move on a modest budget if the physics is right, and its band, now opened, will never close again. The chapters that follow tell how the uncatchable light was caught, and what it showed when it was.

Chapter 1: The Discovery of NuSTAR

The mission has a prehistory, the decades of the unfocusable band, and the prehistory begins with the physics of the mirror. The X-rays, whether soft or hard, will not reflect head-on: the light's wavelength is so short that it penetrates the surface it strikes, and the only reflection is the grazing one, the ray skimming the mirror at fractions of a degree, like a stone skipping water. The soft X-rays could be made to skip: the great observatories' mirror shells, polished to atomic smoothness and nested in dozens of cylinders, caught the soft band at its shallow angles, and Chandra's and XMM's images made the technique famous. The hard band was another matter: at the higher energies, the grazing angle shrinks further than any single surface can serve, and the old mirrors' reflectance fell to nothing, the hard light passing through the optics as if they were absent. For forty years the band's observers flew the collimators, the mechanical baffles that carved the sky into broad patches, and the coded masks, the pinhole cameras of the field, instruments that could locate a source but never image it with the mirror's power.

The mission's enabling idea, the multilayer mirror, arrived by materials science. The depth-graded multilayer, a coating of alternating thin films, tungsten and silicon or platinum and carbon, laid down in layer after layer of precisely varied thickness, works as a Bragg reflector for the grazing X-ray: the stacked layers reinforce the reflection at the angle the single surface refuses, and a mirror coated with the right grading can reflect the hard band at angles ten times wider than the bare metal allows. The invention's pioneers, in the X-ray optics laboratories of the 1990s, measured the coatings' promise on the bench; the mission's concept, first proposed and deferred in the small-explorer competitions of that decade, won its start in 2003, the instrument chosen in the fifth round of NASA's small-explorer selections, and the phase of construction began.

The discovery of NuSTAR, in the sense of the mission made real, belongs to its team and its tests. The principal investigator, Fiona Harrison of Caltech, the physicist who led the concept through the competitions and the construction; the two mirror modules, built in the partner laboratories, each holding over a hundred nested shells, the innermost coated for the middle band and the outer for the hardest; the focal-plane detectors, the pixel arrays of cadmium zinc telluride, a semiconductor that stops the fierce photons the older detectors let pass; and the ten-meter mast, the longest fixed structure ever flown as part of an astronomical telescope, the focal length the optics demanded and no fairing could house, the mast's deployable design the mission's boldest engineering. The tests, the vibration and the thermal and the deployable's trials, ran for eight years of construction, and the mission's own personnel called the telescope's story the long road from the bench to the pad.

A further thread of the prehistory deserves its paragraph, the failed predecessors and the long road to approval, the mission's own near misses. The focused hard band was proposed repeatedly before it flew: the band's concept missions of the 1990s, the graded-optics proposals that the review boards declined, the fiscal era's cancellations that removed one approved mission of the class before its construction began, the telescope's own first selection, deferred in the budget's cut and rebuilt for the later round; the road's lesson, the mission's own historians note, is the explorer program's own pattern, the good idea kept alive by its champions through the lean rounds until the fiscal weather cleared. The reader should hold the thread's meaning: NuSTAR was not the first focused hard-band

telescope proposed, only the first one launched, and the band's opening waited on a decade of paperwork as much as on the coating's physics, the mission's true start date sitting somewhere in the declined proposals the record preserves.

The construction years, the mission's own building story, deserve their own paragraph in the discovery's account. The optics, the mirror shells' polishing and the coatings' laying-down, the months of the sputtering chambers' work, each shell's grade tuned and tested; the detector work, the cadmium zinc telluride's pixel arrays, the cryostat-free design the room-temperature sensors allow, the focal bench's own instrument culture; the mast's procurement, the truss's space-heritage and the mission's adaptation of it, the deployment test's own rehearsals; and the assembly's whole period, the two optics modules and the focal bench mated to the spacecraft bus, the environmental tests' gauntlet, the vibration and the vacuum and the deployments' rehearsals again: the years' story is the small mission's own economy, the team's hands on the hardware through the whole chain, the lab-bench culture carried to the launch pad. The thread's meaning for the reader is the discovery's own texture: the first focused hard-band image was eight years of bench-work and test-chambers in the making, and the mission's true long road runs from the sputtering chamber to the Pegasus's drop, the whole account of this chapter's ending standing on the construction years' quiet accumulation.

One more thread belongs in the discovery chapter, the competition's own story, the small-explorer selection rounds the mission survived. The NASA selection process, the announcements of opportunity and the proposal reviews, the field's whole community competing for the few slots: the mission's own submissions, the concept's first proposals and the boards' verdicts, the selections won and the budget's cancellation that removed the first approval, the rebuilding for the later round; the competing proposals of the same rounds, the other bands' missions, the selection's own odds the record keeps; and the final award, the fifth small-explorer round's verdict, the mission chosen and funded in the early two-thousands: the thread is the mission's institutional biography, and the meaning carried is the selection's own weight, the explorer slots so scarce that a field's whole decade can hang on one review's ranking, and the hard band's opening, delayed once by the fiscal weather and won once by the proposal's persistence, is the survivor's story as much as the physics's, the record of how the impossible band finally got its telescope.

One further thread of the origins deserves recording, the detector team's own history, the heavy-semiconductor's long road. The cadmium zinc telluride of the focal plane was a material with a biography of its own: the compound's early promise, the room-temperature hard-photon stopping the silicon could not do, the material-science decades of growing the crystals and taming the contacts; the detector community's patient work, the pixel arrays' slow maturation in the laboratory era of the 1990s and 2000s, the space qualification's own proving; and the mission's own adoption, the arrays' performance finally reaching the focal-plane standard, the room-temperature detectors that the small mission's power budget could host: the thread is the sensor's own origin story, and its inclusion completes the mission's three-part inheritance, the coatings, the mast, and the crystals, each technology arriving at the pad on its own long road from the bench. The reader should hold the pattern's own meaning: the telescope was possible when three separate maturations happened to finish at once, the optics and the structure and the sensors converging on the same decade, the mission the meeting point of three histories, and the band's forty-year wait ending not in one invention but in the coincidence of three, the focused hard light requiring the whole set, and the set completing at last.

The launch-day record deserves one more paragraph, the first hours' own drama, the deployment watched. The Pegasus drop and the ignition, the rocket's own short arc to orbit, the spacecraft's separation and its first signals; the mast's extension, the mission's boldest single act, the structure's lengthening commanded on the first day's operations, the team's confirmation of the ten-meter geometry, the whole mission's optical design verified in the one command's success; and the first checkout weeks, the detectors' activation and the optics' own first looks, the calibration targets' early stares, the instrument's health assessed instrument by instrument: the record is the mission's own birth certificate, and its drama is the engineering's, the years of construction hanging on the hours' events, the small mission's whole life resting on a truss that lengthened and a detector that woke. The paragraph's meaning is the account's own continuity: every focused image the mission has ever taken stands on the launch morning's one quiet success, and the reader who knows the morning knows the whole harvest's foundation, the universe's hardest light's first eye opened over the Pacific in the minutes after a rocket fell from a plane.

The launch completed the discovery, and its date belongs in the record: the thirteenth of June, 2012, the Pegasus rocket, the small launcher dropped from the belly of an aircraft over the Pacific near Kwajalein, the telescope falling away and firing into orbit, the mast extending to its full length in flight, and the first light arriving a month later, the

picture of the black-hole system Cygnus X-1, the band's first focused image, the unfocusable light focused at last. The discovery of NuSTAR, in sum, is the story of a forty-year-old problem solved by a coating and a mast, and the reader who wants the mission's whole meaning in one sentence holds it: the universe's hardest light was made to skip, and the skipping made pictures.